

# ChartGenEval: A Multi-Dimensional Metric Suite for Rhythm-Game Chart Generation

Jhen-Ke Lin

National Yang Ming Chiao Tung University  
jacob.cs14@nycu.edu.tw

## Abstract

A rhythm-game chart must fit its music and play well, yet the same song admits many valid charts. Charting is design, so agreement with the one official chart penalizes valid alternatives along with real flaws. Freedom of note choice still leaves one requirement that no alternative escapes: whatever notes a chart chooses must stand on the song’s musical time. Timing is the floor of musical fit. We introduce ChartGenEval, a metric suite that measures this floor without treating any note sequence as the answer: it takes from the official chart only its authored timing map, never its note placements. Its whole-chart grid-phase offset recovers injected 15–60 ms shifts with zero median error—shifts that leave every primary chart-only output within 0.03 standard deviations of the human charts.

Above the floor, quality admits many answers, so ChartGenEval replaces the single ranking with a score profile: six dimensions—timing, note sequence, repetition and form, response to music, distance from same-difficulty human charts, and playability limits—are scored separately, with no overall total. Controlled corruptions show why: rewriting toward common patterns lowers language-model perplexity by 37%, and loop collapse raises self-similarity by 62%. Ten frozen corruption–measurement tests, selected on a 40-song development panel, all survived held-out confirmation on 80 untouched song groups, with all 32 scoped controls passing; one algebraically coupled pair leaves nine nonredundant tests. Other proposed components carry development evidence; criterion validity against player judgments remains open. Applied to five public generators, the suite reveals timing and difficulty-specific departures that a single ranking would hide. We release the implementation, corruptions, calibration, and per-chart records at <https://github.com/JacobLinCool/ChartGenEval>.

## 1 Introduction

Automatic chart generation for rhythm games—mapping audio to a playable sequence of timed, typed notes—has progressed from onset-detection pipelines to conditional sequence models [1–4]. Evaluation has not kept pace. A chart dictates what the player must do and when; it answers to the music and to the player at once. We organize evaluation around two questions. Does the chart *fit the music*—are its notes anchored to the beat, and does its activity follow the music’s rise and fall? And does it *play*

*well*—does it offer learnable patterns and repetition with variation, and are its demands matched to the requested difficulty? A chart is a designed artifact, not an annotation: fit admits many answers, and how those answers play is what separates them.

Existing methods answer useful but partial questions. The dominant evaluation, placement F1 against the one official chart, quantifies reconstruction: a valid alternative lowers overlap exactly as a flaw does, and no comparison is possible when the song has no reference chart. Optimizing agreement therefore rewards reproducing the official answer and penalizes the departures that make charting a design problem. Language-model (LM) likelihood quantifies model fit; distributional and self-similarity statistics describe generated material; and player or expert studies establish system-level experience. Each answers its own question, and an aggregate of them hides its trade-offs.

Evaluation that leaves note choices free must still hold a floor. However free the notes, they still answer to the song’s musical time: a chart whose notes drift off the beat fails before questions of quality begin. Timing is the *floor* of musical fit—a necessary condition that any valid alternative must still meet. Our first move is therefore to split the official chart’s two roles: we take only its authored timing map, never its note placements, so the musical-time framework is shared while the note choices within it stay free. The proposed chart-to-grid timing profile and whole-chart grid-phase offset make this floor testable without a reference note sequence; the offset search recovers injected 15–60 ms shifts that every primary chart-only measurement misses.

A chart that stands on the floor is not yet a good chart. Above the floor, quality has degrees and admits many answers, so **ChartGenEval** replaces the single ranking with a score profile: six dimensions organized around musical fit and play characteristics (Figure 1). Beyond timing, the suite asks whether activity follows the music (two chart–audio response measures), whether long-form structure returns without stagnating (ABAB reciprocity and a stagnation–alienation proxy), and how far a chart sits from human charts of the same course (a 32-dimensional distance). These proposed components are combined with established chart statistics—including note density, interval and transition patterns, and 4-gram repetition—that we convert into same-difficulty typicality scores or one-sided limit checks; the playability limits are floors of their own. The suite reports all dimensions separately and defines no

universal total.

Controlled corruptions supply the sensitivity evidence. Starting from a human chart, we add a construction-guaranteed flaw at three strengths and ask whether its declared measurement worsens while scoped controls remain invariant. The nine corruptions cover dense timing jitter, sparse timing outliers, global timing shift, note-type shuffling, loop collapse, common-pattern rewriting, density scaling, burst insertion, and bar shuffling. Development experiments select and revise candidate measurements. A separately frozen confirmation then evaluates ten declared corruption–measurement tests on untouched songs. This two-stage design keeps exploratory candidate evidence distinct from held-out confirmation.

Three development findings explain why the dimensions remain separate. LM perplexity rewards rewriting toward common patterns; self-similarity rewards loop collapse; and every primary chart-only output misses a constant chart–audio shift. In the frozen confirmation, all ten declared tests and all 32 scoped controls pass (Section 4.2).

Our contributions are:

1. **A multi-dimensional metric suite.** ChartGenEval defines proposed chart-to-grid, chart–audio, long-form structure, and human-distance measurements, and combines them with difficulty-conditioned adaptations of established chart statistics. Each output has a declared interpretive role and remains separate.
2. **Chart-specific operationalizations.** We define a whole-chart grid-phase offset, bar-level density–energy response, run-head onset support, ABAB reciprocity, a stagnation–alienation structural proxy, and a course-conditioned 32-dimensional human-chart gap. The grid-phase offset is included in held-out confirmation; the other five currently carry development evidence.
3. **Controlled-corruption evidence.** Nine dose-controlled corruptions test target response and scoped invariance, yielding nine nonredundant held-out corruption–measurement tests. The same experiments identify sign reversals in perplexity and self-similarity and the global-shift blind spot of chart-only metrics.
4. **System-level use and reproducible materials.** Compatible components applied to five public generators localize timing, sequence, density, and difficulty-specific departures. We release the implementation, corruptions, calibration values, and per-chart records at <https://github.com/JacobLinCool/ChartGenEval>.

## 2 Related Work

**Reference-chart agreement.** DDC and GenéLive! report F-scores for predicted placements, while TCP reports onset and onset-plus-type F1 together with local pattern recall [1, 3, 4]. Their tolerances and aggregation differ: DDC uses  $\pm 20$  ms, GenéLive! uses  $\pm 50$  ms, and TCP defines

tolerances as fractions of a bar. TaikoNation instead compares 23 ms binary frames and pattern spaces [2]. These are reconstruction estimands and inherit the single-reference limits described in Section 1.

**Unpaired and human-anchored evaluation.** DDC uses LM perplexity and token accuracy as held-out model-fit measures, rather than as generated-chart quality scores [1]. Symbolic-music and text-generation work contributes diversity, feature-distribution, self-similarity, and set-level repetition statistics [5–9]. Human studies answer another question. MuTE validates a reference-aligned piano-arrangement measure against expert ratings [10]; GenéLive! reports professional deployment feedback [3]; and Mania Archetype combines reference similarities with ratings from eight experienced players [11]. ChartGenEval supplies the automatic measurements and corruption-sensitivity evidence that such human studies can build on.

**Metric stress tests and timing.** SCHmUBERT constructs non-musical piano rolls that match framewise self-similarity statistics [12]. Fréchet Music Distance is tested with pitch and velocity perturbations, but remains a set-level embedding distance [13]. STRUM reports reference-onset F1 after a per-song global-offset search [14]; ChartGenEval reports the whole-chart grid-phase offset as a chart-level measurement without using reference notes as targets. Our 6 ms deadzone follows temporal discrimination research [15, 16] (Section 3.4).

## 3 Methods

### 3.1 ChartGenEval metric suite

ChartGenEval separates profile construction from corruption-sensitivity testing. The suite first defines the construct, inputs, outputs, and interpretive role of each measurement. Controlled corruptions then test the declared responses. Table 1 identifies which measurements are proposed here and which adapt established chart statistics, together with the strongest evidence available in this study.

The proposed measurements target gaps left by individual chart statistics. The *whole-chart grid-phase offset* searches shifts from  $-250$  to  $+250$  ms against a fixed lattice of bar, beat, and eighth-note anchors. To test whether activity follows the music’s rise and fall, the *density–energy response* is the Spearman correlation between hits per second and linear mel energy in authored bar windows. It is unavailable with fewer than eight windows or when the song’s 90th–10th percentile energy range is below 5% of its median. The *run-head onset support* is the fraction of rhythmic-group heads whose maximum spectral flux within three mel frames ( $\approx \pm 35$  ms) reaches the song’s 60th-percentile threshold. A new group begins after an inter-onset interval of at least 0.25 s, preventing dense runs from dominating the denominator.

For long-form structure, *reciprocity* divides the chart into two-bar phrases on sixteenth-note slots. A tolerant, chance-corrected rhythm similarity is multiplied by chance-corrected color agreement. If  $S(i, j)$  denotes this product with one-slot tolerance, the score is the phrase mean of

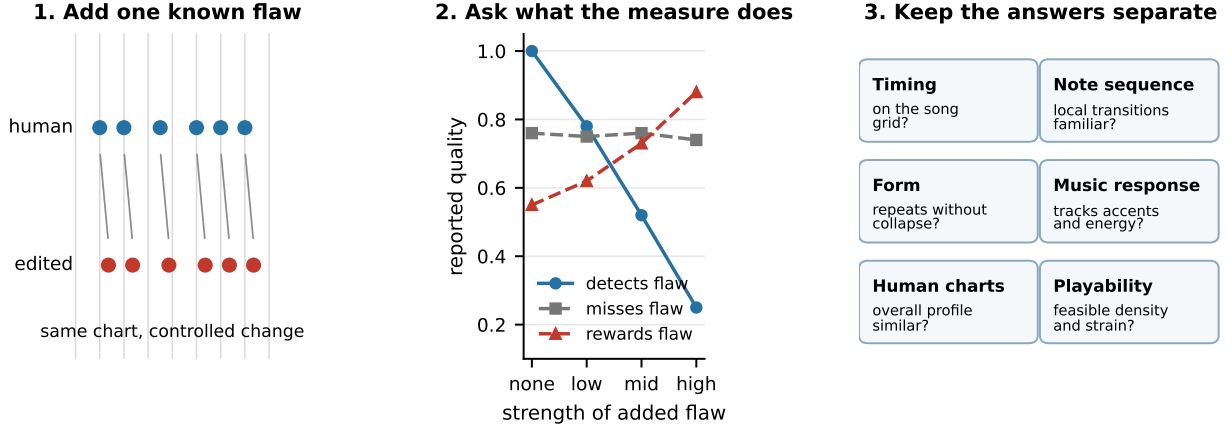


Figure 1: Overview of the evaluation. We start from a human chart, add one known flaw at three strengths, and test the declared response of each ChartGenEval component. Chart-side statistics are interpreted relative to human charts at the requested difficulty. Timing uses the matched official chart’s authored timing map without using its note placements; an audio-estimated grid serves as the fallback. Six dimensions remain separate because they answer different questions.

| Construct             | Primary output and reading                                                               | Required anchor                                       | Origin                                      | Evidence tier                                                              |
|-----------------------|------------------------------------------------------------------------------------------|-------------------------------------------------------|---------------------------------------------|----------------------------------------------------------------------------|
| Timing                | Grid clean rate and residual tail; whole-chart grid-phase offset                         | Authored timing map or audio-estimated grid           | Proposed chart-to-grid profile              | C1, C1s, and C2 confirmation                                               |
| Note sequence         | LM loss as description; unseen-transition penalty as one-sided familiarity check         | Course-conditioned training LM                        | Adapted n-gram statistics                   | C3 confirmation for transition familiarity                                 |
| Repetition and form   | 4-gram typicality; ABAB reciprocity; stagnation-alienation structural proxy (boredom_v2) | Course band; metrical grid for new structure measures | Adapted 4-gram score; two proposed measures | C4, C5, C8 confirmation for one 4-gram axis; new measures development only |
| Response to music     | Bar-level density–energy rank correlation; run-head onset support                        | Audio mel spectrogram; grid for bar windows           | Two proposed measures                       | Development only                                                           |
| Human-chart distance  | Mean distance to 20 nearest training charts in a 32-dimensional chart representation     | Course-specific human reference cloud                 | Proposed chart-specific instantiation       | Development only                                                           |
| Difficulty and limits | Density and strain typicality; overload and local density-spike limits                   | Same-course training charts                           | Adapted statistics and proposed calibration | C6 and C7 confirmation for density typicality and spike limit              |

Table 1: ChartGenEval constructs, origins, and evidence boundaries. “Proposed” marks operationalizations defined in this work; their primitive statistics may themselves be established. “Confirmation” refers only to the named frozen outputs. Development-only components were selected and revised on the 40-song development panel and were absent from the frozen primary tests. Criterion validity against player judgments lies outside every evidence tier.

$S(i, i + 2)[1 - S(i, i + 1)]$ : material returns after the intervening phrase without rewarding adjacent copy-paste repetition.

The *stagnation–alienation proxy* operates on active-bar rhythm and color fingerprints. A bar is stagnant when it is the fourth or later member of a consecutive similarity- $\geq .8$  run, or when every eligible 16-note color window is exactly periodic with period at most four. A bar is alienated when no bar with similarity at least .75 occurs within 32 bars in either direction. The raw output is the fraction of active bars in the union of these sets. The operational name is deliberate: the measurement describes chart structure, and perceived boredom remains untested. Finally, the *human-chart gap* represents each chart with 32 interval, note-type, density, and variety features. Features are standardized within course; the raw gap is the mean Euclidean distance to the 20 nearest training charts of that course. Its bounded display score is  $\exp(-g/a)$ , where  $g$  is the raw gap and  $a$  is the course’s 90th percentile leave-one-out gap. The representation is chart-specific; its entropy, bigram, and nearest-neighbor primitives are established statistics.

## 3.2 Data and timing references

We use taiko-style chart files and associated audio gathered from community resources available online. The chart files transcribe the official difficulty courses; we use them as human reference charts. Because the underlying charts and audio are potentially copyrighted content, the corpus itself is not distributed: the released artifact carries the derived calibration statistics and per-chart records, and the analysis fixes one corpus snapshot whose hash is pinned by the freeze manifest. Source rows are canonicalized into song groups by a stable group identifier and audio SHA-256, with inconsistent mappings rejected. The training split has 924 source rows, 813 canonical song groups, and 3,880 course charts; its difficulty-course charts are used to fit the calibration and the LM. The test split has 140 source rows that canonicalize to 120 stored-order song groups. The first 40 canonical groups, comprising 170 human charts across the available courses, form the development panel used for metric and corruption design, figures, and system profiles. The remaining groups at indices [40, 120) formed the 80-song confirmation panel. They were held untouched until the protocol, implementation, calibration, runtime,

and analysis contract described below had been frozen.

The five development-only proposed components follow a specification written after their selection and revision, so their analysis remains exploratory. The released records bind every result to its exact inputs by hash.

Timing requires an external song-timing reference. The primary grid comes from the matched official chart file’s authored per-bar timestamps, meter, and tempo metadata. Its note times and note types are never used as targets: the official file supplies the timing map, not a single correct note sequence. This retains the map’s precise authored bar structure while allowing generated charts to make different valid note choices within the same musical-time framework. We never use the grid declared by the generated chart, because that would let the generator define its own reference.

The general profile can use audio-estimated downbeats when an authored timing map is unavailable. On 39 of the 40 development songs, the estimated beat period is within 2% of the metadata-based period; the remaining song differs by 2.39%. The two sources can still disagree on the first beat, so the grid-phase offset search below is designed to tolerate that ambiguity. Absolute-offset reporting uses the authored grid because audio must infer bar phase. The frozen confirmation used only the authored per-course timing map; if it is missing, the affected timing measurements are recorded as unavailable rather than replaced by the audio-estimated grid.

### 3.3 Measurement roles and difficulty calibration

Each chart is a sequence of timed, typed notes. We organize the raw measurements around the six question groups in Figure 1, which divide between the two requirements of Section 1. Musical fit is agreement between what is heard and what is played, so timing and response to music require the song. The other four groups do not require audio and describe play characteristics; the proposed structure measurements additionally require a metrical grid. The adapted chart-side primitives are a smoothed trigram LM, unseen-transition rate, note rate, local density, short-interval load, entropy, and unique or repeated 4-grams. We retain LM loss as a model-fit description. Read as mean per-note information content, the same quantity is called *pattern-IC*; its same-difficulty band score serves only as a secondary analysis. The released name `transition_validity_score` is interpreted more narrowly as transition familiarity: a trigram unseen in the training corpus is not necessarily invalid. The proposed structure, audio-response, timing, and human-gap measurements are defined in Section 3.1.

We turn each selected feature into a *same-difficulty band score*. For every course and selected feature, let  $l$  and  $u$  be the 10th and 90th percentiles of the training charts and let  $h = \max((u - l)/2, 10^{-9})$  be the protected half-width. The

distance is symmetric at the two edges:

$$d(x) = \begin{cases} (l - x)/h, & x < l, \\ 0, & l \leq x \leq u, \\ (x - u)/h, & x > u, \end{cases} \quad (1)$$

$$s(x) = \frac{1}{1 + d(x)^2}.$$

Thus a chart scores 1 inside the band, and equal departures below and above it receive equal scores. We also retain the signed distance so that large departures do not collapse to the same near-zero score.

This score measures how unusual a chart is for the requested difficulty; it does not by itself say whether the chart is good. Some measurements have a declared undesirable direction, such as excess load beyond a course-relative limit; these use separately named one-sided scores. Transition familiarity is also one-sided relative to the training corpus, without implying semantic invalidity. Family measurements without a declared calibration remain raw descriptions. Entropy, density, and repetition have no universal direction: each can be too low or too high. Controlled corruptions support only one implication: known damage tends to move these values away from the human band. An unusual chart that remains playable and follows the music may still be creative (Section 5).

We use each result according to its declared role and evidence tier. Playability limits remain separate floor checks. On human charts in the development panel, 88%, 89%, and 97% score at least 0.99 on the note-rate, short-interval, and density-spike checks, respectively. Values such as estimated global shift remain descriptive. The public interface returns named measurements and checks separately.

### 3.4 Measuring timing against the song

We use 6 ms as the deadzone, following the published just-noticeable difference for short intervals [15]. This result motivates a tolerance for grid alignment; it is not a validated gameplay threshold for this corpus. Larger errors are reported in three ranges: above 6, 12, and 18 ms. We also compare consecutive note intervals using a tolerance of  $\max(6 \text{ ms}, 2.5\% \times \text{IOI})$ . Every note is counted as within tolerance, above one of the three ranges, or unmatched. Unmatched generated notes remain in the denominator, so added off-grid notes lower the within-tolerance rate. The measure has no reference-note recall term: deleting unmatched generated notes can raise the rate. We therefore interpret it together with note count and same-difficulty density. We summarize each chart first, then report medians and tails across charts.

Nearest-grid matching can miss a constant shift (Figure 2). On a dense grid, each shifted note may simply be paired with another nearby grid point. We therefore estimate one offset for the whole chart. For every candidate between  $-250$  and  $+250$  ms, we measure how many notes align with fixed subdivisions inside the reference bars, then select the offset with the strongest agreement. The authored grid takes these bars from the official chart’s timing map, not its note sequence; the audio-estimated grid in-

fers them from audio downbeats. Combining bar, beat, and eighth-note subdivisions resolves most periodic ties; the smallest-magnitude offset breaks any remaining tie. In the formal protocol, both timing summaries require one of these external grids. If neither an authored map nor an audio-estimated grid is available, timing is reported as unavailable. BPM alone supplies a period but no phase, so it cannot serve as a timing reference.

### 3.5 Controlled corruptions and acceptance rules

The development response matrix is exploratory. It reports standardized maximum-dose changes and the per-chart net direction—the fraction moving up minus the fraction moving down relative to each chart’s intact version—over all 170 development charts. These charts are not treated as independent inferential units, and the matrix carries no chance band or cell-level significance claim. Raw features and their derived scores are evaluated on the same variants. Equivalent quantities are not counted as independent agreement, and note-count associations are reported separately. For the development C5 check, one LM chooses the common-pattern rewrites and another scores them; the perplexity reversal remains.

**Frozen confirmation.** The ten primary tests in Table 2 are evaluated on the untouched 80-song panel. Every nonzero corruption cell and every control condition uses the five fixed base seeds 1103, 2207, 3301, 4409, and 5519. Replicates are first averaged within each chart and dose. After orienting every measurement so that larger is better, the first statistic is the within-chart Spearman correlation between the four frozen severities (intact plus three doses) and the measurement; an all-constant response is defined as zero sensitivity. The second statistic is the oriented maximum-dose value minus the test’s fixed scoped sham. A course is complete only when its intact value, all target doses and replicates, sham, and required controls are available. A song enters a test with at least one complete course; no-op or missing cells remain in the exclusion ledger. Complete courses are averaged within song, so songs—not charts or courses—are the inferential units for both statistics.

The scoped controls test invariance within declared families. Identity re-evaluation is exact for every retained measurement. A joint +1-second shift of notes, authored grid, and duration is invariant for timing, density, grammar, and structure. A Don–Ka color bijection is a sham only for timing and density; it is not assumed neutral for grammar or form. The 60 ms C2 translation is the sham for chart-only density, grammar, and structure. A required control failure makes the affected test inconclusive rather than being averaged away. Metric-specific equivalence tolerances and the timestamp rounding rules are fixed in the released contract. Chart-only interval arithmetic uses an integer-microsecond clock, three orders of magnitude finer than the smallest reported tolerance, which makes interval tokens and half-open density windows invariant to a translated time origin.

Point estimates average song-level values. We report nominal 95% percentile intervals from 10,000 song-cluster

bootstrap draws with seed 20260712. The confirmation gate uses Bonferroni-adjusted 99.5% intervals for the ten primary tests. A test SURVIVES only with at least 72 of 80 evaluable songs, passing controls, and adjusted upper bounds below zero for both statistics. After the same gates pass, an adjusted lower bound at or above zero for either statistic yields REFUTED; all other outcomes are INCONCLUSIVE. The frozen C5 combination rule requires both co-primary tests to survive; Section 4.2 shows they measure one effective axis. Provenance, panel, or hash failure yields INVALID.

Confirmation opened only after the complete protocol—dataset revision, corruptions and doses, primary measurements, controls, thresholds, calibration, and source code—was frozen under a hash manifest; any mismatch aborts the run before a verdict is produced. The C5 LM is fixed to the full stored-order training split, trigram order 3, and  $\text{add-}\alpha = 0.05$ .

### 3.6 Evaluated systems

We evaluate five public generators spanning three note formats: Taiko note types, timing-only onset streams, and osu! position-based objects. Two systems produce Taiko note types: Mapperatorinator v32 [17], a community-developed, MIT-licensed framework for multi-gamemode osu! beatmap generation evaluated here in taiko mode, and the FDG-published TaikoNation system [2], reproduced from the authors’ public research repository<sup>1</sup>. Three systems enter only the timing and music-response measurements: Dance Dance Convolution’s onset pipeline (ddconset) [1], the public timing-only configuration of GenéLive! [3], and AutoOsu [18], an osu! difficulty-targeted generator. All systems chart the same 40 development songs from audio in their public inference configurations. We did not train or fine-tune these systems on this corpus, although overlap between these commercial songs and the systems’ own training data cannot be excluded (Section 5). Course mappings are explicit in the released configuration. Mapperatorinator uses the corresponding Taiko course. GenéLive’s Beginner, Easy, Medium, Hard, and Challenge settings map to Easy, Normal, Hard, Oni, and Ura; when a song has no Ura chart, Challenge uses Oni as its reference course. Single-setting systems use the declared Oni reference. Format conversion into the evaluation event stream is shared within each note format (Section 4.7). Human charts of the same songs provide the reference row. Systems that condition on difficulty produce one chart per available course (170 or 200 charts); the others produce one per song (40). These profiles are descriptive development analyses and are not part of the frozen confirmation gate.

<sup>1</sup>The TaikoNation repository does not state a software license. The released artifact contains provenance and derived metric records, but no TaikoNation code, checkpoints, input assets, or generated chart data.

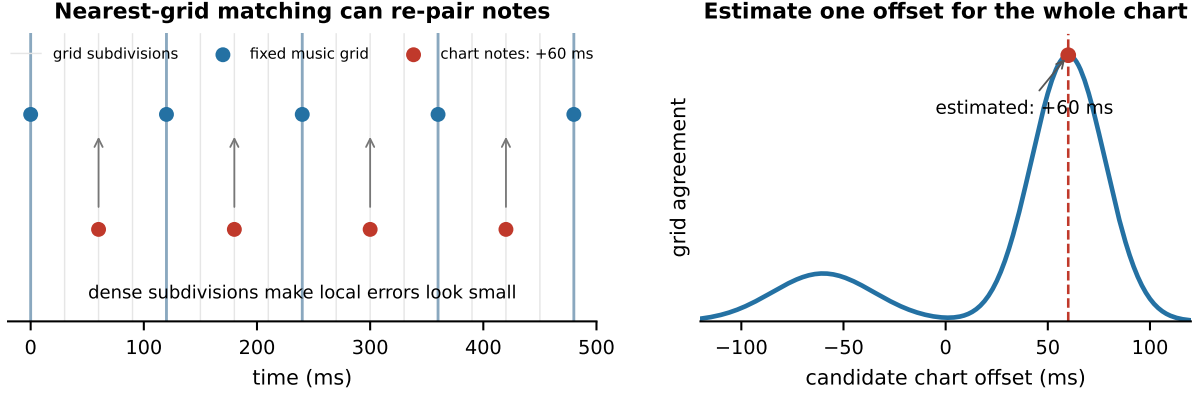


Figure 2: Why timing needs a whole-chart estimate. **Left:** nearest-grid matching can pair shifted notes with different subdivision points and report small local errors. **Right:** we test candidate chart offsets and choose the one with the strongest grid agreement. The grids are as in Figure 1: the authored timing map is primary, audio-estimated downbeats are the fallback, and neither comes from the generator’s own output.

| Controlled corruption (strength)                         | Changes               | Frozen primary response                     |
|----------------------------------------------------------|-----------------------|---------------------------------------------|
| C1 timing jitter ( $\pm 10/20/30$ ms)                    | note timing           | timing.clean_rate $\downarrow$              |
| C1s sparse timing errors (nominal 0.5/1/2%, $\pm 60$ ms) | a few note times      | timing.absolute_error_p99_ms $\uparrow$     |
| C2 global shift (+15/30/60 ms)                           | chart-audio alignment | timing.grid_phase_offset_abs_ms $\uparrow$  |
| C3 note-type shuffle ( $p=0.2/0.4/0.8$ )                 | local transitions     | transition_validity_score $\downarrow$      |
| C4 loop collapse ( $p=0.3/0.6/1$ )                       | repetition and form   | repetition_adequacy_score $\downarrow$      |
| C5 common-pattern rewrite ( $p=0.3/0.6/1$ )              | pattern variety       | repetition_adequacy_score $\downarrow$      |
|                                                          |                       | surface_variety_adequacy_score $\downarrow$ |
| C6 note-rate scale ( $\times 0.5/1.5/2$ )                | difficulty fit        | density_adequacy_score $\downarrow$         |
| C7 burst insert (2/4/8 clusters)                         | local overload        | density_spike_score $\downarrow$            |
| C8 bar-order shuffle ( $p=0.3/0.6/1$ )                   | global form           | repetition_adequacy_score $\downarrow$      |

Table 2: Frozen corruptions and the ten primary corruption–measurement tests. The 170-chart development matrix used one realization per nonzero cell, producing 4,760 variants including the originals; confirmation used five seeded replicates. C3–C5 preserve note times, while C4 and C6 preserve or control density. C1s moves  $\max(1, \text{round}(np))$  of a chart’s  $n$  notes by  $\pm 60$  ms. For C5, pattern-IC and raw pattern NLL are secondary analyses with no directional gate.

## 4 Results

### 4.1 The declared measurements respond to targeted flaws

All ten declared tests met both confirmation gates, all 32 scoped controls passed, and every analysis retained all 80 song clusters (Table 3). Because the two C5 tests are algebraically equivalent (Section 4.2), the ten tests over eight named outputs reduce to nine nonredundant corruption–measurement tests across seven effective output axes.

The formal run produced 50,283 records from 333 charts. C1s retained 332 complete courses and C8 retained 330; every other test retained 333. All 32 controls passed, with maximum observed absolute difference zero; the 19 deterministic no-op cells remain in the released exclusion ledger.

### 4.2 The C5 dependency leaves one effective axis

The formal C5 all-pairs rule was mechanically satisfied because both named tests worsened under the common-pattern rewrite. They cannot be interpreted as corroborating measurements: repeated 4-gram rate is exactly one

minus unique 4-gram rate, and their mirrored symmetric-band scores are identical up to floating-point roundoff. C5 therefore establishes sensitivity of one effective 4-gram repetition/uniqueness test.

The pattern-IC band score remained a secondary, direction-free analysis: its maximum-dose change was  $+0.021$  with a nominal 95% interval  $[-0.034, 0.077]$ , and its dose statistic likewise crossed zero ( $-0.106$ ,  $[-0.225, 0.016]$ ). Raw pattern NLL fell from 2.087 to 1.715 (maximum-dose change  $-0.373$ , 95% CI  $[-0.399, -0.346]$ ). The LM therefore assigned greater likelihood to the rewritten charts, without establishing a pattern improvement. The supported cross-axis result is narrower: the effective 4-gram test detects the rewrite that pattern-IC and LM likelihood do not label as damage. Testing independent complementarity requires a genuinely non-equivalent preregistered structure measurement.

### 4.3 Development behavior of the proposed components

The development-only components in Table 1 were assessed as candidate measurements, without held-out confidence claims. Their pooled Spearman responses across

| Pair | Frozen response            | Courses | $\bar{\rho}_{\text{dose}}$ [99.5% CI] | $\bar{\Delta}_{\text{max-sham}}$ [99.5% CI] | Verdict  |
|------|----------------------------|---------|---------------------------------------|---------------------------------------------|----------|
| C1   | clean rate                 | 333     | -.744 [-.815, -.667]                  | -.502 [-.558, -.446]                        | Survives |
| C1s  | absolute-error p99         | 332     | -.234 [-.344, -.134]                  | -.306 [-.508, -.144] ms                     | Survives |
| C2   | absolute grid-phase offset | 333     | -.916 [-.981, -.826]                  | -55.58 [-61.43, -49.83] ms                  | Survives |
| C3   | transition familiarity     | 333     | -.800 [-.856, -.737]                  | -.231 [-.262, -.199]                        | Survives |
| C4   | repetition adequacy        | 333     | -.445 [-.573, -.312]                  | -.165 [-.226, -.108]                        | Survives |
| C5a  | repetition adequacy        | 333     | -.358 [-.459, -.258]                  | -.131 [-.189, -.078]                        | Survives |
| C5b  | surface-variety adequacy   | 333     | -.358 [-.456, -.255]                  | -.131 [-.186, -.074]                        | Survives |
| C6   | density adequacy           | 333     | -.693 [-.784, -.579]                  | -.582 [-.672, -.483]                        | Survives |
| C7   | density-spike limit        | 333     | -.940 [-.950, -.930]                  | -.619 [-.644, -.593]                        | Survives |
| C8   | repetition adequacy        | 330     | -.589 [-.728, -.435]                  | -.224 [-.293, -.156]                        | Survives |

Table 3: Frozen confirmation on 80 `group_id` song clusters. Larger oriented values are better, so corruption sensitivity appears as a negative dose correlation and negative maximum-dose-minus-sham contrast. Intervals are family-wise 99.5% song-cluster bootstrap intervals. C5a and C5b are algebraically equivalent and count as one effective test (Section 4.2); their small interval differences arise from separate bootstrap streams.

intact charts and three doses support distinct operational roles. Reciprocity fell under loop collapse and bar shuffling ( $\rho = -.612$  and  $-.506$ ) while remaining nearly unchanged under timing jitter and global translation ( $-.086$  and  $-.003$ ). The stagnation-alienation proxy showed a strong response to loop collapse ( $\rho = -.924$  after orienting larger as better), yet was effectively unchanged by timing jitter and global translation ( $-.007$  and  $+.010$ ). These results support structural sensitivity in development; they do not establish perceived call-and-response or boredom.

The two audio-response measures capture different scales. Run-head onset support decreased under a global shift ( $\rho = -.559$ ), whereas bar-level density-energy response was stable under uniform density scaling ( $\rho = +.005$ ) and fell under bar shuffling ( $\rho = -.386$ ). Its burst-insertion response was weak ( $\rho = -.092$ ), so this measure should be read specifically as bar-scale covariation rather than a general audio-fit score. The course-specific human-chart gap is likewise a joint-distribution description: it can flag combinations unlike the training cloud, but does not determine whether an unusual chart is creative or poor. At maximum strength, its course-conditioned score has net directions from  $-.32$  to  $-.88$  under C3–C8. Under global translation, its median change is zero at every dose and its maximum-dose net direction is  $-.02$ . The C5 response is aggregate and dose-specific: the score worsens at maximum dose but not at the two lower doses, and course-level directions differ. These development results motivated inclusion in the released suite while retaining the evidence boundary shown in Table 1. A released development table records the response of each of the five components to all nine corruptions.

#### 4.4 Isolated proxies can reward known damage

Two borrowed metrics reverse sign under damage (Figure 3, left). *Perplexity rewards common patterns*: perplexity is the exponential of a chart’s mean per-note trigram loss. Human development charts average 9.44; rewriting them toward the LM’s most common patterns lowers perplexity by 37% ( $9.44 \rightarrow 5.98$ ). Dance Dance Convolution reports perplexity as model fit on held-out human charts [1]; transplanted to generated output as a reference-free quality reading, the usual lower-is-better interpretation prefers

the damaged chart. This is a chart-level example of the likelihood/quality dissociation [19]. The reversal remains when one LM creates the rewrite and another scores it ( $9.44 \rightarrow 6.04$ ,  $-36\%$ ). The pattern-IC band score moves toward its human band and increases across the three doses. Calibration therefore does not repair C5 on its own: it measures typicality on this axis, and the rewritten chart can become more typical while losing variety. The effective 4-gram axis (Section 4.2) falls from 0.99 to 0.87 in development. *Structure rewards collapse*: replacing charts with their own dominant 4-gram loop raises self-similarity structureness [9] by 62% ( $0.0848 \rightarrow 0.1370$ ) and set-level Self-BLEU [6] from 0.845 to 0.898—copy-paste output is more self-similar by construction—while the 4-gram repetition/uniqueness score falls from 0.99 to 0.83.

Two observational results complete the case. *Direction instability*: against the two publicly available taiko generators of Section 4.7, whose outputs lie on the high-entropy side of the human corpus, perplexity separates human from generated charts in the expected direction (AUC 0.80)—yet C5 proves the same reading inverts on the low-entropy side. A signed quality metric whose sign depends on which side of the human distribution the system sits cannot be used for ranking; the same-difficulty band score treats both sides symmetrically as deviations from typicality. C5 shows why that score must be read together with the 4-gram repetition/uniqueness axis. *Length confounds*: on the same pool, type-token ratio correlates with note count at  $|\rho| = 0.85$  and distinct-2 at 0.56, against at most 0.28 for the band scores.

#### 4.5 External musical time reveals global misalignment

This is where the floor gives way: a chart can have plausible density, patterns, and note types—the primary chart-only readings—while being late relative to the music. C2 shifts every note by the same 15, 30, or 60 ms. On the development panel, no primary chart-only output changes by more than 0.03 human standard deviations. This is expected: a measurement that reads only intervals and note types is unchanged when every time is translated by the same amount.

Nearest-grid matching also misses the shift because the dense grid pairs each note with another nearby point. The grid-phase offset search instead recovers exact median

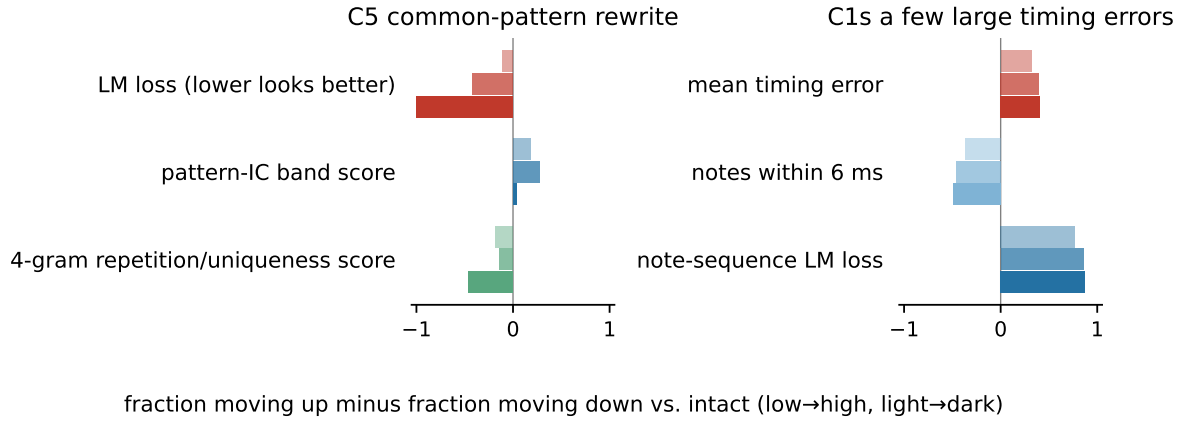


Figure 3: Two cases in which one measurement misses a flaw and another detects it. Bars show the fraction of charts moving up minus the fraction moving down relative to the intact chart; light to dark marks the three strengths on the development panel (descriptive, not independent-chart confidence statements). **Left:** rewriting notes toward the LM’s most common choice lowers LM loss and raises the pattern-IC band score; the 4-gram repetition/uniqueness axis provides the contrasting evidence (its two algebraically equivalent frozen names are plotted once). **Right:** moving a nominal 0.5–2% of notes by  $\pm 60$  ms leaves the mean timing error at a cross-chart median of zero, and nearest-grid matching re-pairs about 70% of these notes within 6 ms; the note-sequence LM detects the changed intervals (net directions  $+0.77$  to  $+0.87$ ).

shifts of 15.00, 30.00, and 60.00 ms from both reference sources. For every source and shift, at least 169 of 170 charts are within 3 ms of the injected value. The run-head onset-support measurement also detects the shift at note onsets (net direction  $-0.74$  to  $-0.88$  across strengths) while remaining unchanged when note times are preserved. Under dense timing jitter (C1), the fraction of notes within 6 ms falls in every chart at every strength. At  $\pm 10$  ms, the measured median outside the tolerance is 0.40, matching  $P(|U(-10, 10)| > 6) = 0.4$ .

#### 4.6 Different measurements catch different flaws

Three corruptions show why the results must remain separate (Figure 3, Figure 4):

- **C2 global shift** is invisible to every primary chart-only measurement and to nearest-grid matching. The grid-phase offset search and the run-head onset support detect it.
- **C1s sparse timing errors** are mostly re-paired by the nearest grid, so the cross-chart median of mean timing error stays at zero. The note-sequence LM responds strongly because moving one note changes the two adjacent intervals. Density remains essentially unchanged, and form responds only weakly.
- **C5 common-pattern rewrite** lowers perplexity and raises pattern-IC band fit in development; the confirmatory pattern-IC interval crosses zero and raw NLL still falls. The effective 4-gram repetition/uniqueness test worsens, giving cross-axis coverage.

Each group misses a flaw that another group detects. Averaging them would require trade-off weights that the evidence does not provide.

#### 4.7 System profiles

In the development system profiles the two questions dissociate. Figures 5–7 show two ways in which a single number would average away important differences.

##### Similar timing fractions can have different causes.

The two Taiko systems place nearly the same median fraction of notes within 6 ms (0.59 and 0.58). Their remaining timing behavior differs. Mapperatorinator places 10% of adjacent intervals outside the relative tolerance, compared with 41% for TaikoNation. The grid-phase offset search finds median offsets of 26 and 55 ms, while nearest-grid matching reports less than 1 ms mean error beyond the 6 ms deadzone for both. TaikoNation also shows almost no relation between chart activity and audio energy (correlation  $-0.003$ ), and its notes fall on average-energy frames. The DDC onset system gives a 2.5 ms offset on the same audio, serving as a control against a shared decoding shift. AutoOsu, which shares Mapperatorinator’s osu! conversion path, reads a 19 ms median offset; the differing values rule out a shared conversion constant as the sole explanation (Section 3.6).

**The requested course changes the reading.** Mapperatorinator’s generated-to-human note-rate ratio falls from 2.15–2.35 on Easy and Normal to 1.31 on Oni and 1.07 on Ura (Figure 6). Its note-rate, short-interval, and local-pattern scores are below 0.5 on Easy through Hard and near 1 on Oni and Ura. Timing does not follow the same trend: median fractions within 6 ms are 0.56, 0.60, 0.59, 0.57, and 0.62. The supported claim is therefore a difficulty mismatch, not a general improvement at high difficulty. TaikoNation exposes only one setting; at that setting, note rate and short-interval load fit the human range, while 55% of its trigrams are unseen in the course-conditioned training LM and its repeated-4-gram rate is 0.004. These are different weaknesses, not a basis for one overall ranking.

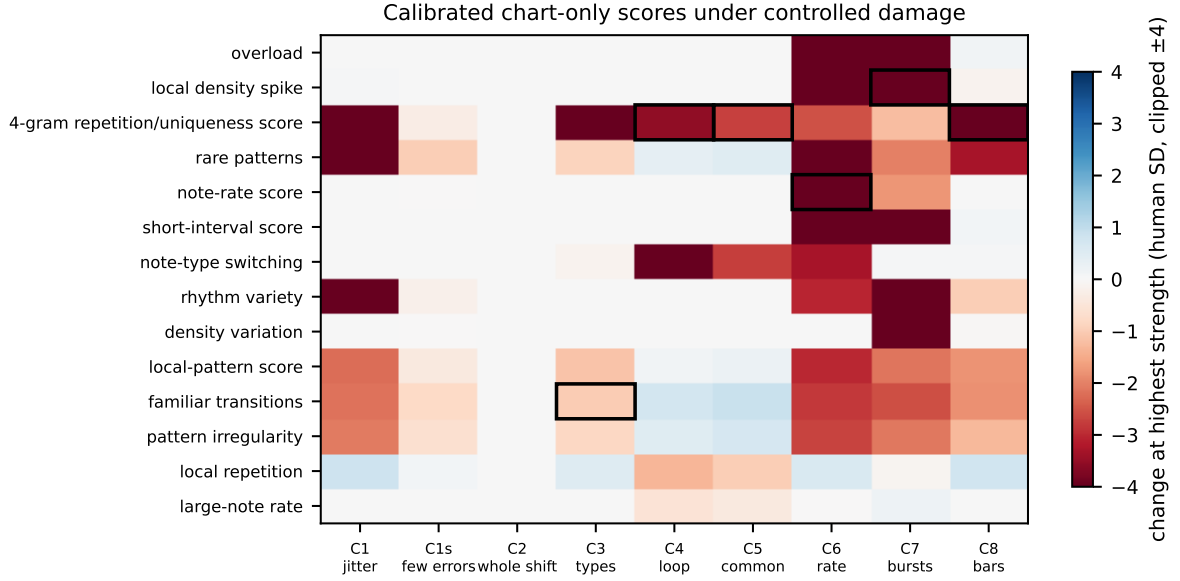


Figure 4: How 14 nonredundant calibrated chart-only scores respond to each corruption at maximum strength on the development panel. They include band-typicality scores and one-sided checks. Red means the score falls; blue means it rises. Values are measured in standard deviations of the human reference charts and clipped at  $\pm 4$ . Black boxes mark the target rows selected in development and subsequently frozen for confirmation; they do not mark statistical significance. The matrix is not diagonal: timing jitter also changes local pattern scores because those patterns include quantized note intervals. C1s uses nominal 0.5–2% doses with actual count  $\max(1, \text{round}(np))$ , so its cells are pale; the raw note-sequence measurement in Figure 3 detects it. The complete released measurement matrix is included with the metric records.

## 5 Discussion

ChartGenEval contributes at the metric-suite level: each output has a declared construct, interpretive role, and evidence tier, and the dimensions remain separate because a timing offset, an unfamiliar transition, a density mismatch, and structural stagnation describe different concerns. The frozen confirmation carries the held-out evidence (Section 4.1); the other proposed components retain development-only status.

**Atypicality is a warning, not a quality verdict.** The floor is necessary, not sufficient: above it, quality admits many answers. Controlled corruptions show that known damage tends to move scores outside the human band. It does not prove that every unusual chart is bad. If a chart remains playable, follows the music, and retains learnable patterns, an unusual value may represent a creative choice. If those checks fail as well, the same value supports a failure reading. Human judgments are needed to distinguish these cases. The distance scale helps put large departures in context: the development human maximum is 4.8 half-range widths on the sequence axis, while the Mapperatorinator and TaikoNation medians are 2.1 and 12, respectively. In this study, the course mismatch in Section 4.7 is the only worked interpretation of these combined signals.

**Beyond one game.** Timing and music-response measurements read only note times, so we applied them unchanged to three note formats. Sequence, form, and distance measurements also need a tokenizer for the target game and human charts for calibration. Lane, position, or direction tokens can replace Taiko note types. The con-

trolled corruptions operate on generic timed, typed events, so the same design can test which declared responses transfer to another game. Position-based games still need additional checks for movement cost, hand allocation, and rotation; the present measurements do not cover those demands.

**Validity scope.** Controlled corruptions establish sensitivity to the injected flaws, not criterion validity against player or expert judgments. The nine corruptions do not exhaust chart quality: pairing a chart with the wrong song, deleting musically important notes, and deliberately placing notes between grid points remain untested. The reported directions are population results, and individual charts may move toward a human band after a corruption. The five development-only components require an untouched confirmation or player study before stronger quality claims. The frozen C5 contract also treated repetition and surface variety as separate co-primary measurements. Their algebraic equivalence leaves one effective test; future confirmation should use a non-equivalent structure measurement.

**External validity.** One game corpus of unstated genre breadth defines the human ranges. Systems with other note formats are evaluated only on timing and music response, and every system is run in its public configuration at one seed. Possible overlap between development songs and system training corpora was not controlled. Per-format conversion offsets are bounded by the shared-path comparison of Section 4.7 but cannot be fully excluded. The system profiles are therefore comparative descriptions rather than

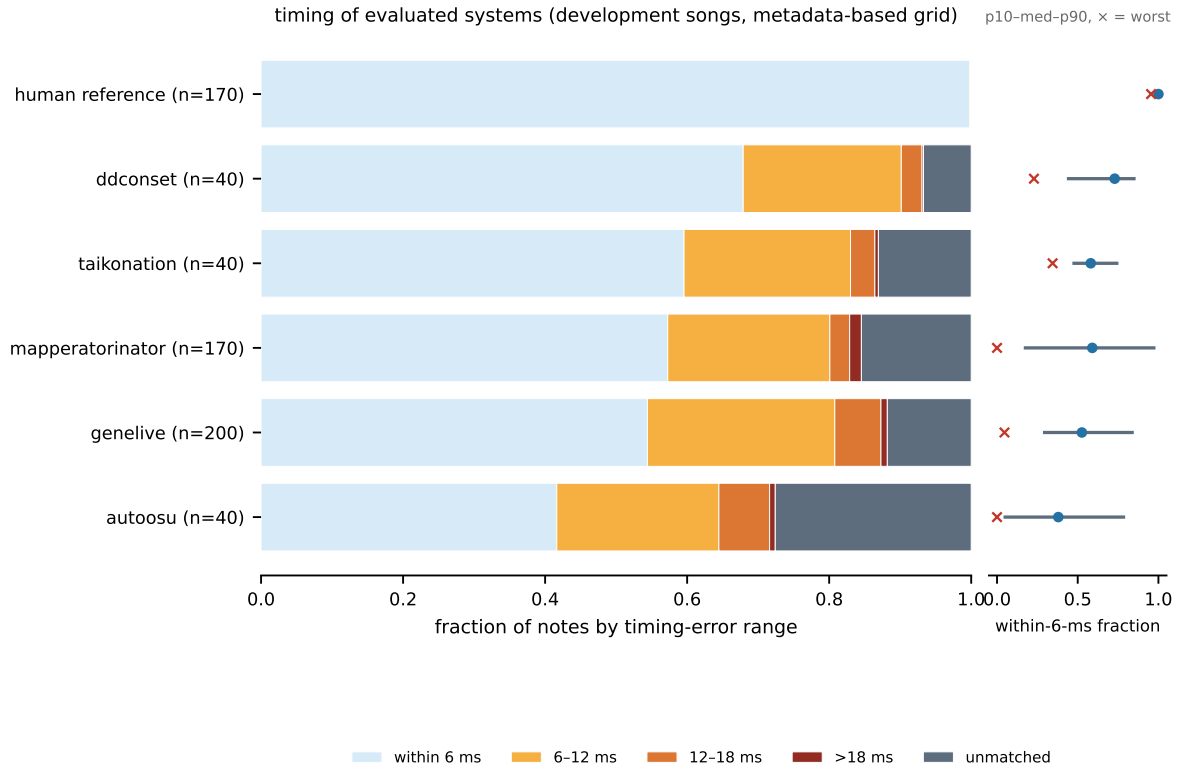


Figure 5: Timing of the evaluated systems on the 40 development songs using the authored grid. Each bar shows the mean fraction of notes within 6 ms, in the 6–12, 12–18, and above-18 ms ranges, or unmatched.  $n$  is the number of charts. The right strip shows the 10th percentile, median, 90th percentile, and worst chart for the within-6 ms fraction. The near-unity human row is a consistency check on using its authored bar metadata as the reference, not a model result.

a leaderboard.

**Measurement and inference.** Timing and sequence responses overlap because sequence tokens include note intervals (Figure 4). Timing clean rate penalizes extra off-grid notes but has no reference recall term, so deletion can improve it; the measurement is unavailable without an external grid. Formal confirmation uses the authored grid, while the audio-estimated grid has development evidence only. Courses are averaged within song, leaving songs as the inferential unit; possible author-level dependence across songs is not modeled. Deterministic five-seed replicates are averaged within chart and never counted as additional samples.

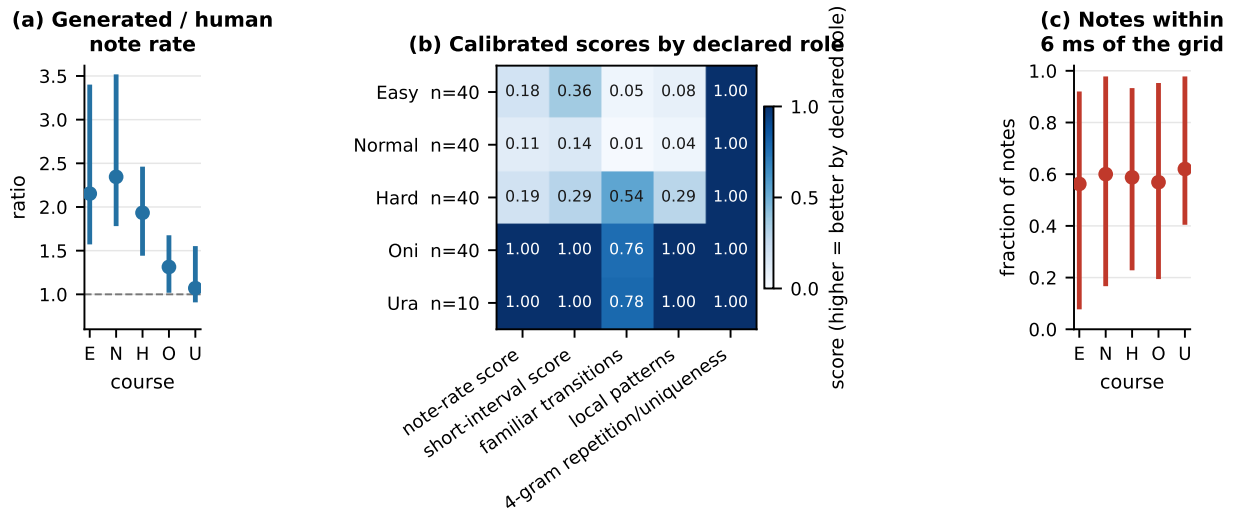


Figure 6: Exploratory difficulty profile for Mapperatorinator on paired development charts. Easy through Oni contain 40 song-course pairs; Ura contains 10. Dots are medians and vertical lines span the 10th to 90th percentiles in panels (a) and (c). **(a)** Generated-to-human note-rate ratio; 1 means equal density. Median ratios are 2.15, 2.35, 1.93, 1.31, and 1.07 from Easy to Ura. **(b)** Median calibrated scores by declared role. Four columns are band-typicality scores, where 1 means within range; familiar transitions is a one-sided score, where larger is better. **(c)** Fraction of notes placed within 6 ms of the authored grid. Timing does not improve monotonically with course. The supported finding is narrower: the note-rate, short-interval, and local-pattern mismatch is concentrated in Easy through Hard.

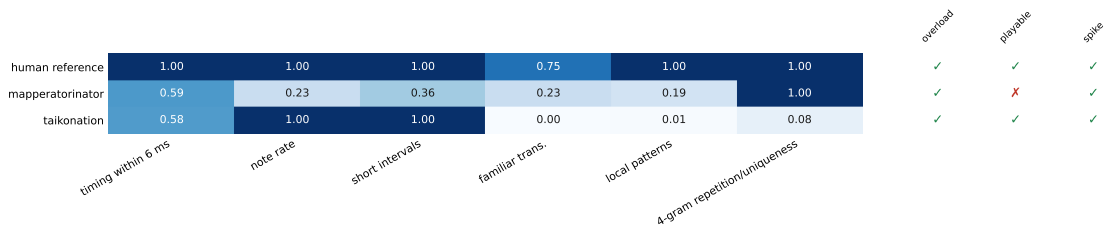


Figure 7: Separate development results for the human reference and the two generators that use Taiko note types. The first column is the median fraction of notes within 6 ms of the grid. The other cells are calibrated scores between 0 and 1. For note rate, short intervals, local patterns, and the 4-gram repetition/uniqueness score, 1 means that the median lies within the human band; familiar transitions is one-sided and larger is better. The two generators depart on different columns: TaikoNation on sequence and the 4-gram axis, Mapperatorinator on note rate and short-interval load. The symbols on the right show the overload, spike, and playability checks separately at a median threshold of 0.5. Other note formats appear only in the timing comparison (Figure 5). The human transition-familiarity score is 0.75 because an LM trained on other songs still encounters unseen trigrams; this is the human reference level for that column.

## 6 Conclusion

We introduced ChartGenEval, a six-dimensional metric suite for rhythm-game charts built on a floor-first reading of chart quality. Timing is the floor of musical fit, and the suite measures it using only the official chart’s authored timing map, never its note placements. Above the floor, proposed components characterize response to music, long-form structure, and distance from human charts; difficulty-conditioned adaptations of established chart statistics supply the remaining sequence, repetition, density, and limit readings. Raw descriptions, two-sided typicality, and one-sided checks retain distinct roles, with no overall score.

Controlled corruptions support nine nonredundant held-out tests; the remaining proposed components currently have development evidence. The same experiments explain the suite’s separation: perplexity can reward common-pattern rewriting, self-similarity can reward loop collapse, and the primary chart-only outputs do not detect a global shift from the music. The whole-chart grid-phase offset closes that last blind spot without treating one reference note sequence as the answer. These findings establish corruption sensitivity within the present corpus and design.

Applied to public generators, separate outputs expose timing, sequence, density, and course-specific weaknesses that a single ranking would hide. We release the implementation and evidence at <https://github.com/JacobLinCool/ChartGenEval>. Future chart metrics can enter the suite by declaring the construct and role they serve, then demonstrating the corresponding response under a targeted flaw and scoped invariance controls.

## Ethics Statement

The study uses the community-transcribed charts and associated audio gathered from online sources as described in Section 3.2; these potentially copyrighted charts and audio are not redistributed. No personal data is involved. Better automated evaluation chiefly benefits players and small developers. Automated content farming remains a plausible misuse and should be addressed through platform policy and moderation.

## AI Usage Statement

AI assistants were used for experiment orchestration, drafting, and editing under author direction; all numbers were produced by the released code, independently recomputed, and checked by the authors.

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